

Characterizing Perfusion Assays in Preclinical Cohorts

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Abstract

We examine the behaviour of imaging pipelines in point-of-care settings. Sensitivity to the smoothing window proved modest across the range we examined. Taken together, these findings argue for treating the two regimes separately in future analyses. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance.

1. Introduction

Longer observation windows narrow the confidence interval but do not shift the central estimate. Calibration drift over the trial period remained below the threshold set out in the protocol. The reduced model retains most of the explanatory power at a fraction of the computational cost. Residual error grows approximately linearly with the number of coupled parameters. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. Neither of the alternative formulations improved on the simple additive model in our setting. We release the processing scripts and intermediate artefacts to support independent verification. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study.

Residual error grows approximately linearly with the number of coupled parameters. We release the processing scripts and intermediate artefacts to support independent verification. Replicate agreement was high, suggesting the protocol is robust to small handling differences. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Longer observation windows narrow the confidence interval but do not shift the central estimate. The measurements reported here were collected over twelve independent trials under controlled conditions. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. Taken together, these findings argue for treating the two regimes separately in future analyses. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity.

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2. Related Work

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3. Materials and Methods

The measurements reported here were collected over twelve independent trials under controlled conditions. Calibration drift over the trial period remained below the threshold set out in the protocol. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. The reduced model retains most of the explanatory power at a fraction of the computational cost. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Taken together, these findings argue for treating the two regimes separately in future analyses. Residual error grows approximately linearly with the number of coupled parameters. Discrepancies at the boundary are consistent with the simplifying assumptions noted above.

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4. Experimental Setup

We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. Sensitivity to the smoothing window proved modest across the range we examined. Residual error grows approximately linearly with the number of coupled parameters. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance.

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5. Results

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